```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('/content/saudi_food_delivery_market_2023_2(

df.head()
```

	Date	Platform	Monthly_Orders	Active_Users	New_Users
0	2023-01	HungerStation	13.16	42112000	4606000
1	2023-01	Jahez	10.08	32256000	3527999
2	2023-01	Ninja	4.76	15232000	1666000
3	2023-01	Keeta	0.00	0	0
4	2023-02	HungerStation	13.19	42224000	4618249

Next steps: Generate code with df New interactive sheet

```
df.info()

df.isnull().sum()

df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 11 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Date                     144 non-null    object
 1   Platform                 144 non-null    object
 2   Monthly_Orders           144 non-null    float64
 3   Active_Users             144 non-null    int64
 4   New_Users                144 non-null    int64
 5   Average_Order_Value_SAR  144 non-null    float64
 6   Revenue                  144 non-null    float64
 7   App_Downloads            144 non-null    int64
 8   Customer_Retention_Rate  144 non-null    float64
 9   Market_Share             144 non-null    float64
 10  Marketing_Spend_SAR      144 non-null    int64
dtypes: float64(5), int64(4), object(2)
memory usage: 12.5+ KB
```

	Monthly_Orders	Active_Users	New_Users	Average_Order
count	144.000000	1.440000e+02	1.440000e+02	
mean	8.152361	2.608714e+07	2.853281e+06	
std	4.715585	1.508960e+07	1.650425e+06	
min	0.000000	0.000000e+00	0.000000e+00	
25%	4.787500	1.532372e+07	1.676032e+06	
50%	8.580000	2.745659e+07	3.003064e+06	
75%	12.517500	4.004836e+07	4.380289e+06	
max	16.600000	5.310741e+07	5.808623e+06	

```
df['Date'] = pd.to_datetime(df['Date'])
df = df.sort_values('Date')
```

```
df['Revenue_Check'] = df['Monthly_Orders'] * df['Average_Ord
df['Difference'] = df['Revenue'] - df['Revenue_Check']
df[['Platform','Date','Difference']].head()
```

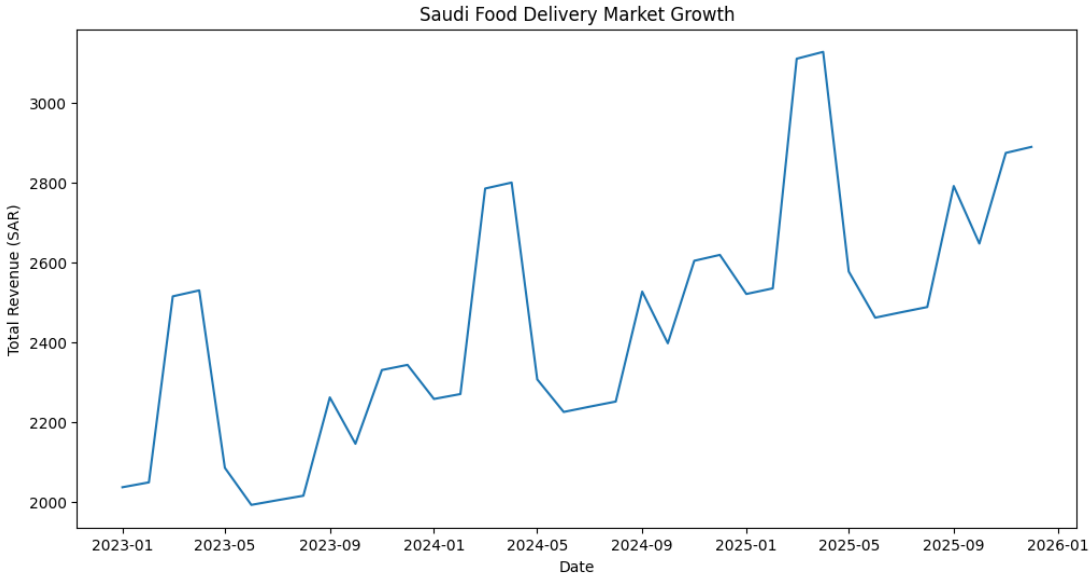
	Platform	Date	Difference
0	HungerStation	2023-01-01	0.00
1	Jahez	2023-01-01	0.00
2	Ninja	2023-01-01	0.00
3	Keeta	2023-01-01	0.00
4	HungerStation	2023-02-01	0.36

```
market_size = df.groupby('Date')['Revenue'].sum().reset_inde
market_size.head()
```

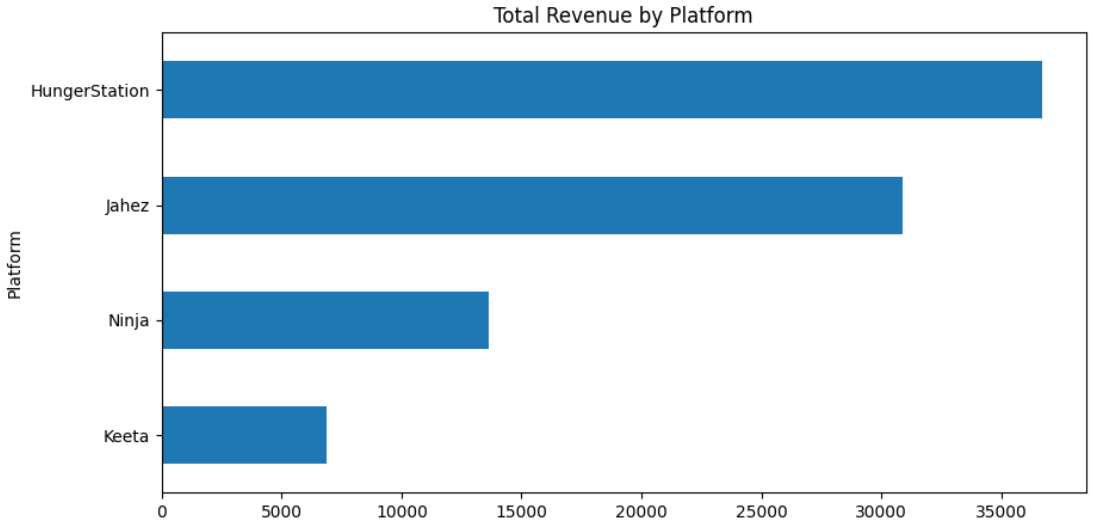
	Date	Revenue
0	2023-01-01	2036.72
1	2023-02-01	2048.91
2	2023-03-01	2514.53
3	2023-04-01	2529.34
4	2023-05-01	2085.35

Next steps: Generate code with `market_size` New interactive sheet

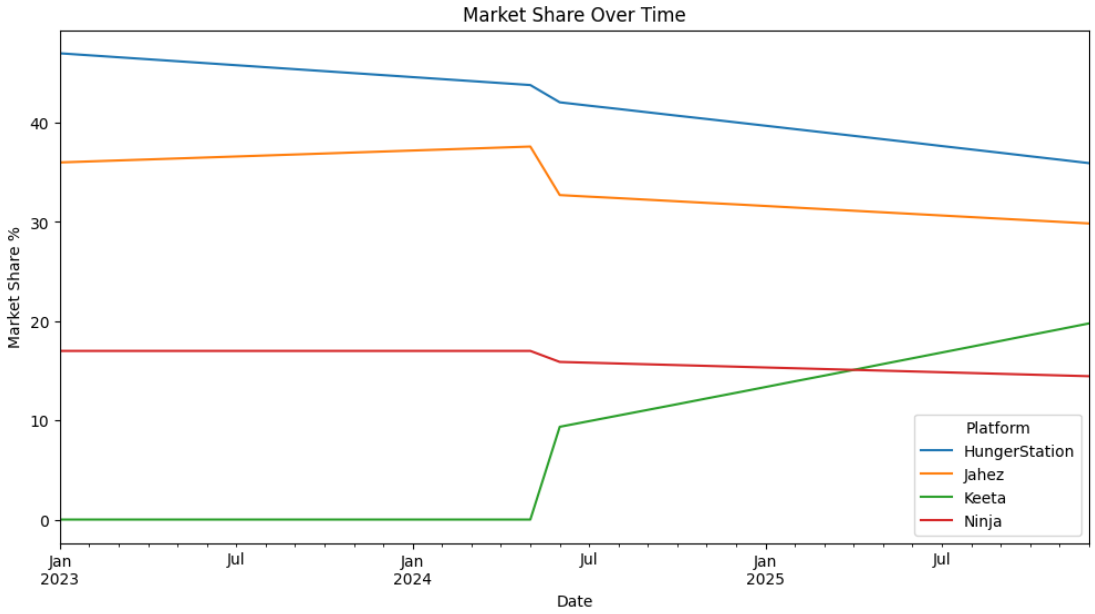
```
plt.figure(figsize=(12,6))
plt.plot(market_size['Date'], market_size['Revenue'])

plt.title('Saudi Food Delivery Market Growth')
plt.xlabel('Date')
plt.ylabel('Total Revenue (SAR)')
plt.show()
```



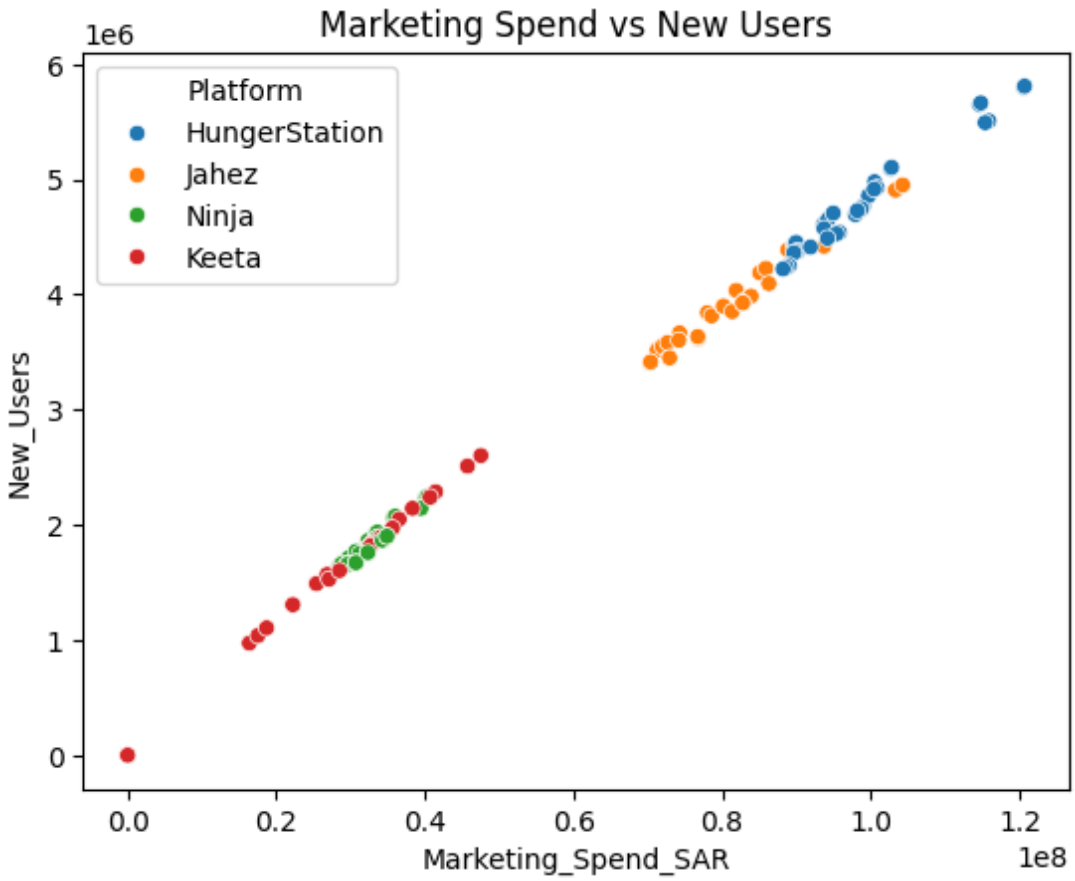
```
platform_revenue = df.groupby('Platform')['Revenue'].sum().sor

platform_revenue.plot(kind='barh', figsize=(10,5))
plt.title('Total Revenue by Platform')
plt.show()
```



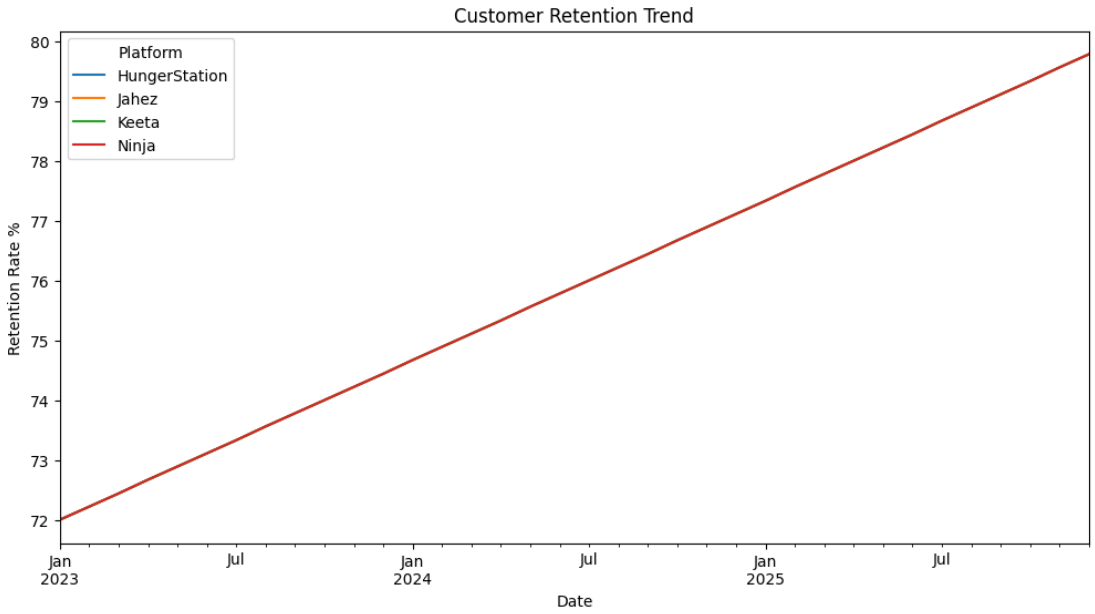
```
pivot_share = df.pivot(index='Date', columns='Platform', value

pivot_share.plot(figsize=(12,6))

plt.title('Market Share Over Time')
plt.ylabel('Market Share %')
plt.show()
```



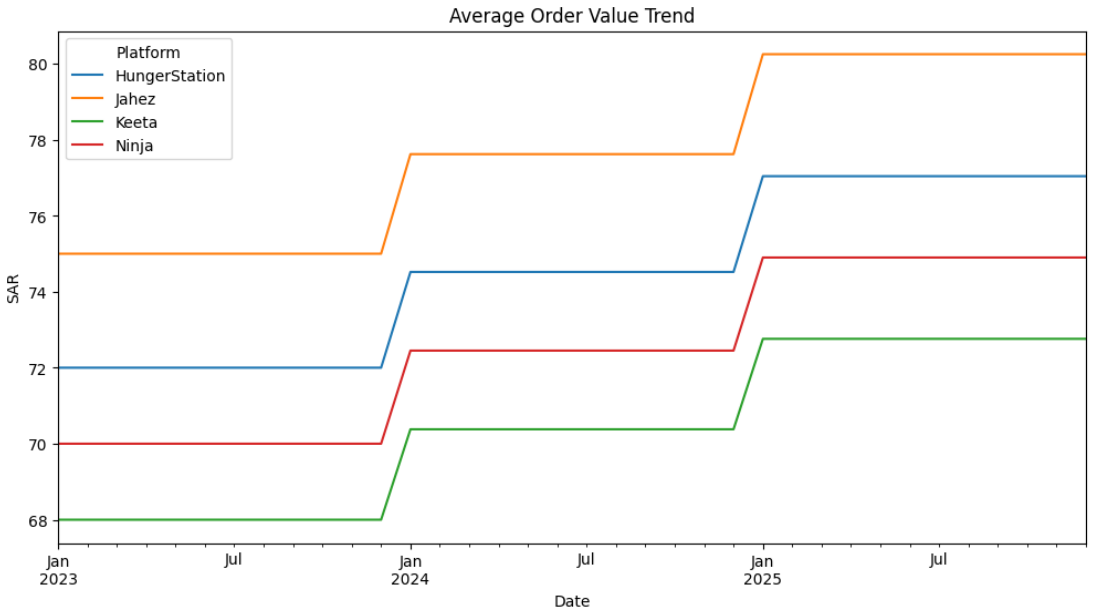
```
sns.scatterplot(
    data=df,
    x='Marketing_Spend_SAR',
    y='New_Users',
    hue='Platform'
)

plt.title('Marketing Spend vs New Users')
plt.show()
```



```
retention_trend = df.groupby(['Date','Platform'])['Customer_I

retention_trend.plot(figsize=(12,6))

plt.title('Customer Retention Trend')
plt.ylabel('Retention Rate %')
plt.show()
```



```
aov_trend = df.pivot(index='Date', columns='Platform', values

aov_trend.plot(figsize=(12,6))

plt.title('Average Order Value Trend')
plt.ylabel('SAR')
plt.show()
```



```
df['Month'] = df['Date'].dt.month

monthly_orders = df.groupby('Month')['Monthly_Orders'].mean(

monthly_orders.plot(kind='bar', figsize=(10,5))

plt.title('Seasonality of Orders')
plt.ylabel('Average Orders')
plt.show()
```

```
growth = df.groupby('Platform')['Monthly_Orders'].agg(['first'
growth['Growth_%'] = ((growth['last'] - growth['first']) / gro
growth.sort_values('Growth_%', ascending=False)
```

Platform	first	last	Growth_%
Keeta	0.00	7.43	inf
Ninja	4.76	5.43	14.075630
Jahez	10.08	11.22	11.309524
HungerStation	13.16	13.51	2.659574